

Morning PRAYER

BEFORE CLASS

Heavenly Father,

Thank You for the gift of this new day
and for bringing us together.

As we begin our class, we ask for Your guidance and blessing.

Grant me, as the teacher, **wisdom, patience, and clarity**
to teach with understanding and inspire my students
to learn with **open minds and willing hearts.**

Bless each student with focus, perseverance,
and the courage to do their best.

May our classroom be filled with **respect,**
kindness, cooperation, and a love for learning.

Guide us in all that we think, say, and do,
so that everything we do may bring
honor and glory to You.

We ask all these through Christ our Lord.

Amen. 

Teach.
Inspire.
Encourage.
Make a
difference.



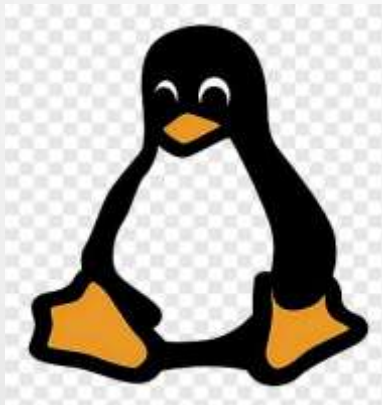
TEACHING with PURPOSE

LEAD • INSPIRE • MOTIVATE

FAITH • HOPE • LOVE

EDUCATION is a WORK
of HEART

Introduction to Operating Systems



BSCS 3109
Operating Systems Configuration and Use

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Learning Outcomes

Knowledge

- ✓ Define an Operating System.
- ✓ Describe the functions of an Operating System.
- ✓ Identify the components of an Operating System.

Skills

- ✓ Differentiate various Operating Systems.
- ✓ Classify Operating Systems according to their purpose.

Learning Agenda

- What is an Operating System?
- History of Operating Systems
 - Functions of an OS
 - Components of an OS
- Types of Operating Systems
- Examples of Modern Operating Systems
 - Interactive Activity
 - Seatwork
 - Quiz

Ice Breaker

imagine...

- What would happen if your computer had:
 - No Windows
 - No Android
 - No Linux
 - No macOS
- Could you open applications?
- Could you browse the internet?
- Could you save files?
- Discussion Question
- "What if there were no Operating System?"

Discussion Question

- Which operating system do you use every day?
- Which OS do you think is the most popular?
- Can you name any other operating systems besides these four?

What is an Operating System?

An Operating System (OS) is system software that acts as an interface between the computer hardware and the user.

It manages

- CPU
- Memory
- Storage
- Input devices
- Output devices
- Software applications

The Operating System is the manager of the entire computer system.

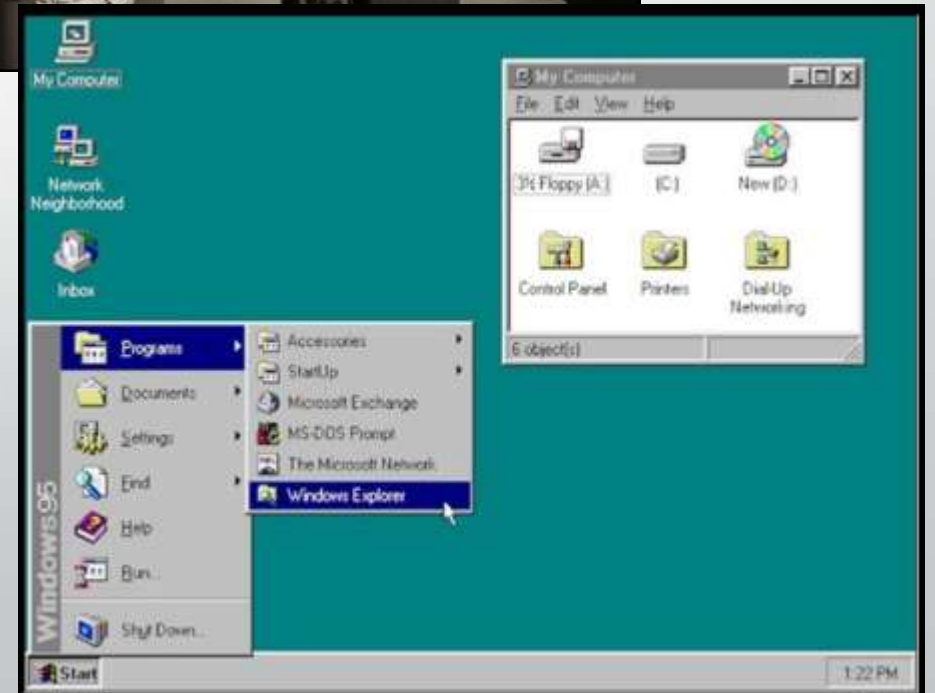
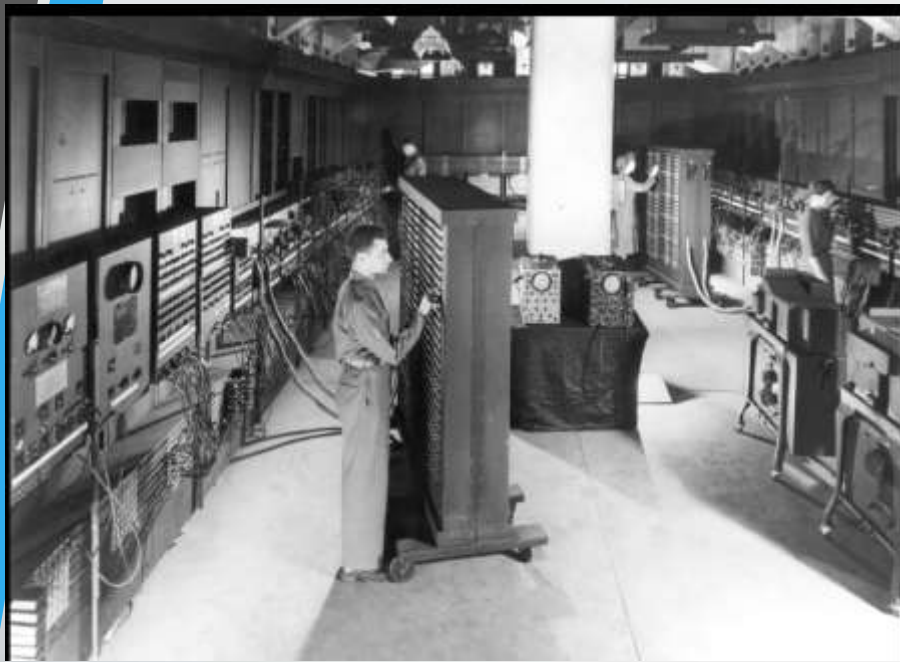
Why is an Operating System Important?

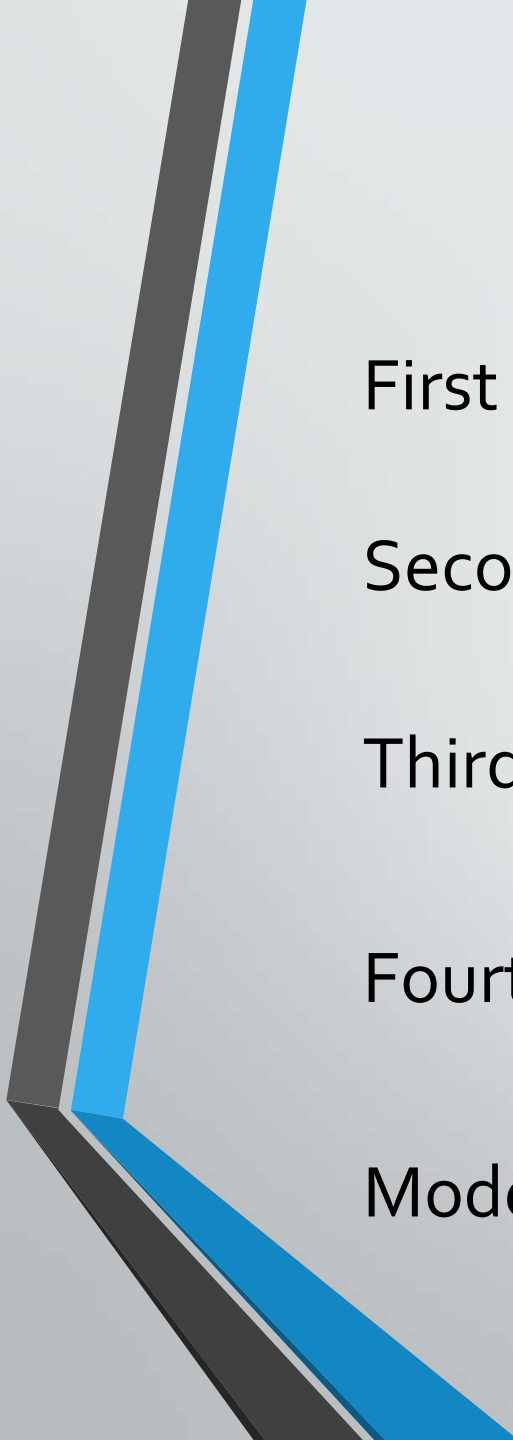
Without an OS...

- ✗ Computer cannot run programs
- ✗ No file management
- ✗ No internet browsing
- ✗ No printing
- ✗ No multitasking

With an OS...

- ✓ Everything works together.





Generation	Period	Major Innovation
First	1940s–1950s	No Operating System
Second	1950s–1960s	Batch Processing
Third	1960s–1970s	Multiprogramming & Time Sharing
Fourth	1980s–Present	GUI-Based Operating Systems
Modern	2000s–Present	Cloud, Virtualization, Mobile OS, AI

Main Functions of an Operating System

The Operating System performs several important tasks.

Process Management

- Runs programs
- Allocates CPU
- Schedules processes



Memory Management

Allocates RAM

Protects
memory

Shares memory

File Management

Stores files

Creates folders

Deletes files

Retrieves data



Device Management

Keyboard

Mouse

Printer

USB

Monitor

Security

User Accounts

Passwords

Permissions

Encryption

Types of Operating Systems

- Batch OS
- Time Sharing OS
- Distributed OS
- Network OS
- Real-Time OS
- Embedded OS
- Mobile OS
- Desktop OS
- Server OS

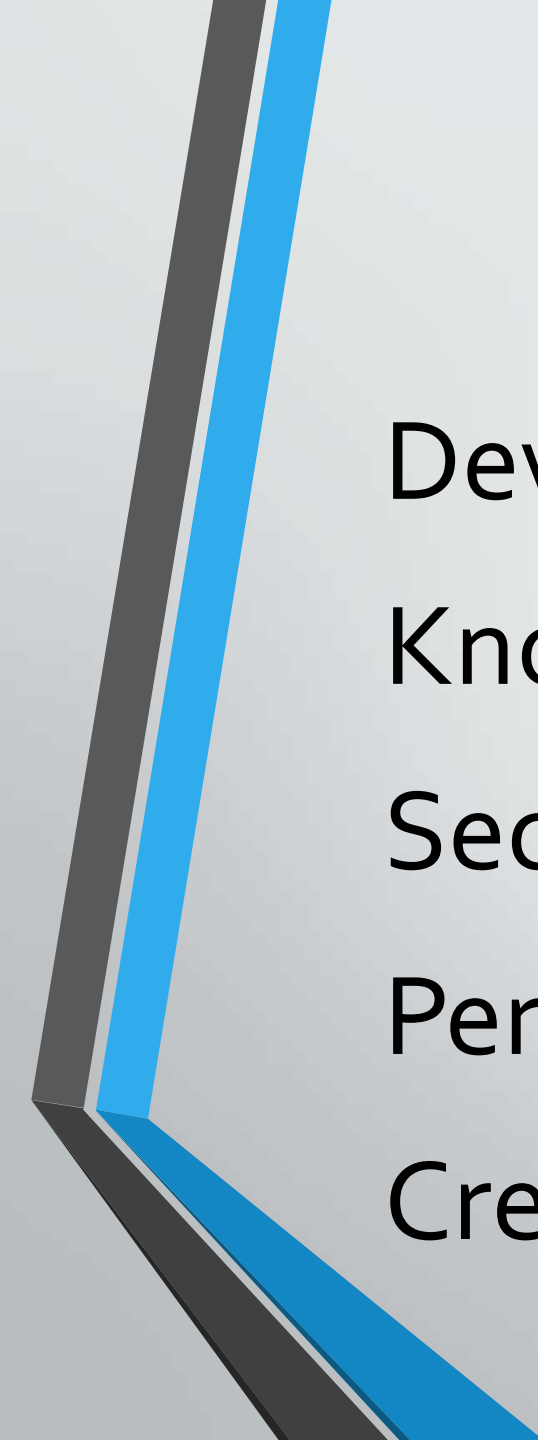
Desktop Operating Systems

Windows

Most popular desktop OS.

Used in

- Schools
- Businesses
- Homes



macOS

Developed by Apple

Known for

Security

Performance

Creativity

Linux

Open Source

Popular among

Servers

Programmers

Cybersecurity

Professionals

Comparison of Popular Operating Systems

Windows

Linux

macOS

Android

Microsoft

Open Source

Apple

Google

GUI

CLI + GUI

GUI

Touch

PC

Server

Mac

Mobile

Group Activity

- What if there were no Operating System?
- Each group discusses
- How would people
 - Save files?
 - Use printers?
 - Play games?
 - Connect to Wi-Fi?

Present your answer in 3 minutes.



THANK YOU!